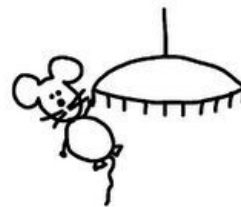


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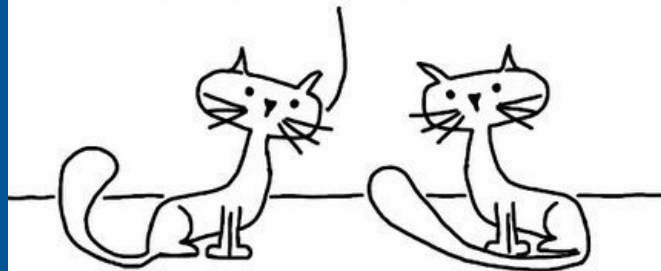
03b - Testability

CPSC310

Nick Bradley & Kyle Chin



I've checked every square foot
in this house. I can confidently
say there are no mice here.



Absence of proof is not proof of absence.

- William Cowper

Learning goals

By the end of the lecture you should be able to:

- Apply the testability axes of controllability and observability to analyze and improve code

Controllability

Definition: the degree to which a test can determine the inputs and state that the code under test depends on.

Lowered by:

- hidden inputs (clock, randomness)
- global or shared state
- hard-wired dependencies
- preconditions that are hard to set up

Observability

Definition: the degree to which a test can inspect the results and effects that the code under test produces.

Lowered by:

- results that are only printed or formatted
- intermediate values that are never returned
- private state that is never exposed
- effects on external systems

Exposing behaviour/state for inspection

```
export class Animal {  
    private hungry: boolean = true;  
    private eaten: number = 0;  
  
    //REQUIRES: Animal to be hungry  
    //EFFECTS:  At the end of the method:  
    //  The animal has eaten more food than it had eaten,  
    //  and the animal is not hungry.  
    public eat(): number {  
        this.hungry = false;  
        this.eaten++;  
        return this.eaten;  
    }  
}
```

It's easier to think of it as a black box first!

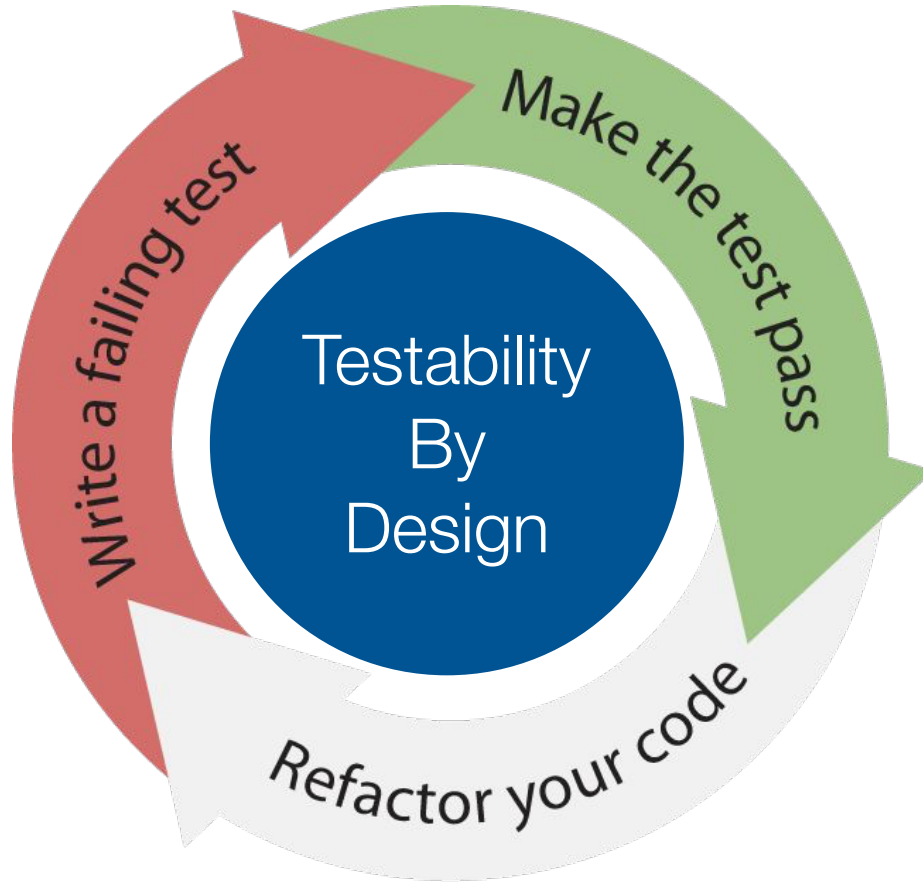
```
describe("Animal", () => {  
  it("should not be hungry and have eaten more food after eating", () => {  
    //Given  
    const a = new Animal();  
    expect(a.isHungry()).to.be.true;  
    expect(a.amountEaten()).to.equal(0);  
  
    //When  
    a.eat();  
  
    //Then  
    expect(a.isHungry()).to.be.false;  
    expect(a.amountEaten()).to.be.greaterThan(0);  
  });  
});
```

Need to add an
isHungry and
amountEaten to the
public interface of
the class



Test-Driven Development (TDD)

Step 1:
Know what
you are
building



Step 2:
details come
from building

Step 3:
Make the
next change
cheap

Testability recap

Code is testable when a test can drive it and check it.

	Controllability	Observability
Question	Can a test choose everything the behaviour depends on?	Can a test see everything the behaviour produces?
Lowered by...	inputs the code finds for itself instead of being given	results the code sends somewhere instead of returning
Raised by...	taking inputs as parameters	returning results as values